



## TRULY INNOVATIVE: TECH

sensor features the fuse spectral images and RGB images to create rich data in dynamic environments and enable advanced computer vision applications. The S1 sensor comes with applicable and optimised optics, a simple interface, a cross-platform API, and the required software stack to yield accurate spectral images. This allows quick integration of the sensor to any application. The sensor will allow designers to display accurate colours in any lighting condition, which was not possible with most existing sensors.

*Spectricity*

<https://spectricity.com>

## NFC wireless charging IC

PTX30W by Panthronics is the industry's first integrated, single-chip solution for the listener circuit in NFC wireless charging systems. It can operate in stand-



alone mode without an external microcontroller to run NFC wireless charging operations. It combines an

efficient rectifier, NFC tag, battery charger and power management, as well as protocol handling. The NFC tag supports bidirectional data communication in NFC Type A mode. This enables the transfer of data between the charging cradle and the charged device, such as the battery's state of charge or fault indicators, as well as enabling firmware upgrades of both devices. The PTX30W will enable designers to build small battery-powered products to implement NFC wireless charging with a board which is around four times smaller than existing designs, thus reducing the size of the product and simplifying the designing process.

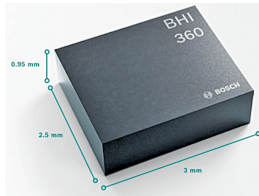
*Panthronics*

<https://www.panthronics.com>

## Programmable smart sensor

BHI360 from Bosch is the smartest programmable inertial measurement unit (IMU) sensor available in the market

today. BHI360 combines a programmable IMU based sensor system that combines a gyroscope with an accelerometer that



allows for full customisation. The IMU comes in a small package

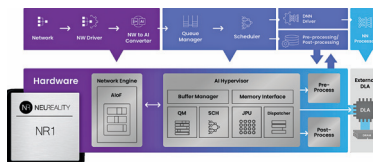
measuring just 3mm × 2.5mm. The integrated sensor fusion library enables 3D audio with head orientation for personalised sound experiences as well as simple gesture recognition. It consists of an ultra-low-power microprocessor, including pre-installed sensor fusion software and algorithms in a single package. It can be used in wearables and hearables, smartphones and tablets, smart devices, etc.

*BOSCH*

<https://www.bosch-sensortec.com>

## Network attached processing unit

NeuReality NR1 is the world's first network attached processing unit (NAPU)—an inference server-on-a-chip with an embedded neural network engine. It is workflow-optimised hardware with specialised processing



units, native network capabilities, and virtualisation capabilities. The NAPU can work with any trained model in any development environment. Its architecture enables inference through hardware with AI-over-fabric, an AI-hypervisor, and AI-pipeline offload. The NR1 will help scientists and engineers to run any AI model faster and easier while reducing the cost.

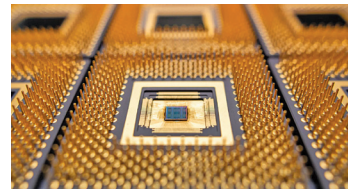
*NeuReality*

<https://www.neureality.ai>

## High-efficiency AI chipset

EnCharge's new computing platform can achieve 150 TOPS/W. This enables the processor to run larger AI models

with higher accuracy, better resolution, and process data from multiple camera channels simultaneously. The test chip can achieve about 20x higher



performance per watt with 14x higher performance per dollar relative to comparable AI accelerators, thus offering revolutionary gains, without time-consuming, cost-prohibitive workarounds. The new computing platform utilises a technique called in-memory computing. The new design can store data and run computation all in the same place on the chip. The EnCharge chipset will thus drastically reduce cost, time, and energy consumption in AI computations.

*EnCharge AI*

<https://enchargeai.com/technology>

## AI edge accelerator module

Axelera AI M.2 is an AI edge accelerator module that is powered by a single Metis AI processing unit (AIPU). It delivers up to 214 tera-operations per second (TOPS) while minimising power consumption and simplifying integration through the downloadable software



stack. The Axelera AI M.2 can provide a significant boost to any vision processing system with next generation form factor (NGFF) M.2 socket. It is a high-performance, cost effective, and user-friendly platform combining hardware and software to accelerate computer vision at the edge. The module provides high processing at a fraction of cost and power consumption of alternative solutions available in the market. It will enable engineers to develop smaller and cost-effective products for computer vision applications.

*Axelera AI*

<https://www.axelera.ai>